Yakeen Plus (2025)

Short Practice Test - 09

DURATION : 60 Minutes | **DATE : 17/11/2024** | **M. MARKS : 192**

ANSWER KEY

PHYSICS		CHEMISTRY		BOTANY		ZOOLOGY	
1.	(2)	13.	(3)	25.	(3)	37.	(2)
2.	(4)	14.	(4)	26.	(2)	38.	(2)
3.	(4)	15.	(3)	27.	(4)	39.	(4)
4.	(4)	16.	(4)	28.	(2)	40.	(2)
5.	(3)	17.	(1)	29.	(4)	41.	(2)
6.	(2)	18.	(1)	30.	(1)	42.	(1)
7.	(4)	19.	(2)	31.	(1)	43.	(2)
8.	(2)	20.	(4)	32.	(4)	44.	(2)
9.	(2)	21.	(4)	33.	(1)	45.	(2)
10.	(2)	22.	(3)	34.	(3)	46.	(1)
11.	(3)	23.	(3)	35.	(3)	47.	(4)
12.	(1)	24.	(1)	36.	(3)	48.	(4)

SECTION-(I) PHYSICS

1. **(2)**

Thresold frequency is maximum for Y and minimum for X.

As we know

$\phi = h\nu_0$

$\phi_Y > \phi_Z > \phi_X$

[New NCERT Class 12th, Page No. 389]

2. **(4)**

Stopping potential of B is greater than A, hence incident frequency of B is greater than A by equation

$h\nu = \phi_0 + eV_0$

[New NCERT Class 12th, Page No. 390]

3. **(4)**

$L = \frac{nh}{2\pi}$

Where n is positive integer.

[New NCERT Class 12th, Page No. 426]

4. **(4)**

D is fissionable.

[New NCERT Class 12th, Page No. 449]

5. **(3)**

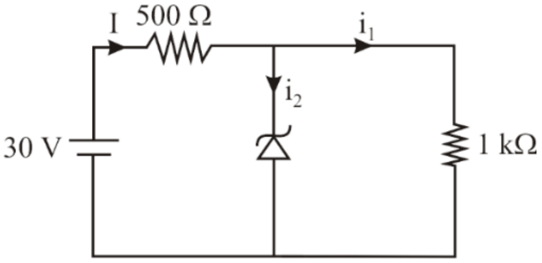
$Y = \overline{\overline{A} + \overline{B}} = \overline{\overline{A}}.\overline{\overline{B}} = A.B$

[New NCERT Class 12th, Page No. 480]

6. **(2)**

Potential drop across dropping resistance = 20 V

Current in dropping resistance $i = \frac{20}{500} = 40\text{mA}$



Current in 1kΩ, $i_1 = \frac{10}{1000} = 10\text{mA}$

Current in Zener diode = (40 – 10) mA = 30 mA

[New NCERT Class 12th, Page No. 472]

7. **(4)**

Only +ve part passes out.

[New NCERT Class 12th, Page No. 484]

8. **(2)**

Density of nucleus is independent of mass number

$\frac{\rho_1}{\rho_2} = \frac{1}{1}$.

[New NCERT Class 12th, Page No. 440]

9. **(2)**

Total B.E. of product = $4x_2$

Total B.E. of reactant = $4x_1$

Energy released Q = $4(x_2 - x_1)$

Q = $4(x_2 - x_1)$

[New NCERT Class 12th, Page No. 447]

10. **(2)**

Binding energy of $\left({}_1^2\text{H} + {}_1^3\text{H}\right) = a + b$

Binding energy of ${}_2^4\text{He} = \text{c}$

In a nuclear reaction, the resultant nucleus is more stable than the reactants. Hence, binding energy of ${}_2^4\text{He}$ will be more than that of $\left({}_1^2\text{H} + {}_1^3\text{H}\right)$

Thus, energy released = charge in binding energy

$= c - (a + b) = c - a - b$

[New NCERT Class 12th, Page No. 452]

11. **(3)**

BE/Nucleon higher then stability is higher hence a heavy nuclei with less *BE*/Nucleon will break in to two lighter nuclei having higher *BE*/Nucleon.

[New NCERT Class 12th, Page No. 441]

12. **(1)**

Lyman series → Electron jumps from higher state to first orbit (n = 1)

Balmer series → Electron jumps from higher state to second orbit (n = 2)

Paschen series → Electron jumps from higher state to third orbit (n = 3)

Pfund series → Electron jumps from higher state to fifth orbit (n = 5)

[New NCERT Class 12th, Page No. 442]

SECTION-(II) CHEMISTRY

13. **(3)**
In Iodoform test the mechanism involves the addition of amine to iodo carbene, no alpha hydrogen should be present to the amine group, and otherwise amine formation takes place. But ethylamine & diethylamine both have alpha hydrogen. Thus, both will give negative iodoform tests
[New NCERT Class 12thPage No. 271]

14. **(4)**
Following tests help to distinguish these:
(i) CH_3OH does not react with Br_2 water. Phenol reacts with bromine water to give white ppt. of 2, 4, 6-Tribromophenol.
(ii) Phenol gives violet colour with $FeCl_3$ whereas CH_3OH does not.
(iii) Phenol (acidic) turns blue litmus red while CH_3OH does not.
[New NCERT Class 12th Page No. 205]

15. **(3)**
The correct match of the experiment with their inferences is given in option (3).
[New NCERT Class 12th Page No. 216]

16. **(4)**
To differentiate between cyclohexane and cyclohexene, we can use bromine water or Bayer's reagent.
[New NCERT Class 12th Page No. 206]

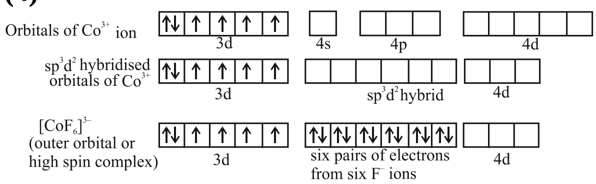
17. **(1)**
The Victor Meyer test is a chemical test used to distinguish between primary, secondary, and tertiary alcohols.
[New NCERT Class 11th Page No. 211]

18. **(1)**
$[Cr(NH_3)_4Cl_2]Cl$ ionizes to give one Cl^- ion. So, it gives 1 mole of AgCl, on the addition of $AgNO_3$.
[New NCERT Class 12th Page No. 119]

19. **(2)**
$[Pt(NH_3)_3(Br)(NO_2)(Cl)]Cl$
Triamminebromidochloridonitroplatinum(IV)chloride
[New NCERT Class 12th Page No. 124]

20. **(4)**
Ma_4b_2 complex does not show optical isomerism.
[New NCERT Class 12th Page No. 127]

21. **(4)**



Number of unpaired electrons = 4
Oxidation state of Co in $[CoF_6]^{3-}$ is +2
[New NCERT Class 12th Page No. 129]

22. **(3)**
Zn, Cd and Hg are the elements of group 12 and show $(n-1)d^{10}ns^2$ electronic configuration.
[New NCERT Class 12th Page No. 90]

23. **(3)**
$\mu = \sqrt{n(n+2)}$ B.M.
where n = number of unpaired e^-.
Both Mn^{2+} and Fe^{3+} have 5 unpaired e^-.
[New NCERT Class 12th Page No. 105]

24. **(1)**
Gd – [Xe] $4f^7 5d^1 6s^2$.
[New NCERT Class 12th Page No. 119]

[Short Practice Test-09 | Yakeen Plus (2025) | 17/11/2024]

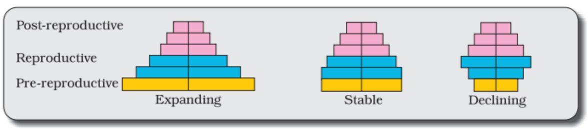
SECTION-(III) BOTANY

25. **(3)**
The formula for logistic growth is equation of

$$dN / dt = rN\left(\frac{K-N}{K}\right).$$

[New NCERT Class 12th Page No. 195]

26. **(2)**



[New NCERT Class 12th Page No. 191, 192]

27. **(4)**
Primary productivity depends on the plant species inhabiting a particular area. It also depends on a variety of environmental factors, availability of nutrients and photosynthetic capacity of plants.
[New NCERT Class 12th Page No. 207]

28. **(2)**

Mutualism	+, +
Parasitism	+, –
Commensalism	+, 0
Amensalism	–, 0

[New NCERT Class 12th Page No. 197]

29. **(4)**

- In an aquatic ecosystem producers include phytoplankton, algae and higher plants.
- Ecosystem need a constant supply of energy to synthesise the molecules they require, to counteract the universal tendency toward increasing disorderliness.

[New NCERT Class 12th Page No. 209]

30. **(1)**

A considerable amount of GPP is utilised by plants in respiration. Gross primary productivity minus respiration losses (R), is the net primary productivity (NPP).

GPP – R = NPP

[New NCERT Class 12th Page No. 207]

31. **(1)**

Warm and moist environment favour decomposition whereas low temperature and anaerobiosis inhibit decomposition.

[New NCERT Class 12th Page No. 208]

32. **(4)**

According to the International Union for Conservation of Nature and Natural Resources (IUCN) (2004), the total number of plant and animal species described so far is slightly more than 1.5 million.

[New NCERT Class 12th Page No. 217]

33. **(1)**

At the ecosystem level, India, for instance, with its deserts, rain forests, mangroves, coral reefs, wetlands, estuaries, and alpine meadows has a greater ecosystem diversity than a Scandinavian country like Norway.

[New NCERT Class 12th Page No. 217]

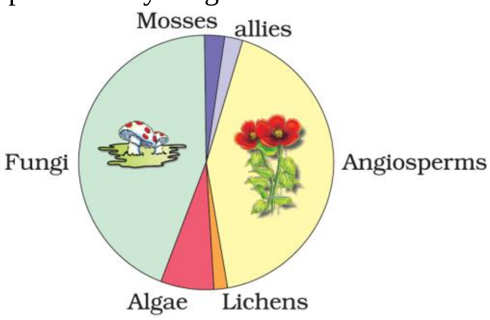
34. **(3)**

Although India has only 2.4 per cent of the world's land area, its share of the global species diversity is an impressive 8.1 per cent.

[New NCERT Class 12th Page No. 219]

35. **(3)**

The highest number of species in the world is represented by fungi.



[New NCERT Class 12th Page No. 218]

36. **(3)**

Three subspecies (Bali, Javan, Caspian) of tiger got extinct.

[New NCERT Class 12th Page No. 221]

SECTION-(IV) ZOOLOGY

37. **(2)**

The Indian Parliament has recently cleared the second amendment of the Indian Patents Bill, that takes such issues into consideration, including patent terms emergency provisions and research and development initiative.

[New NCERT Class 12th Page No. 185]

38. **(2)**

- Allele frequencies in a population are stable and is constant from generation to generation.
- Sum total of all the allelic frequencies is one.

[New NCERT Class 12th Page No. 121]

39. **(4)**

Biotechnology essentially deals with industrial-scale production of biopharmaceuticals and biologicals using genetically modified microbes, fungi, plants and animals.

[New NCERT Class 12th Page No.184]

40. **(2)**

Transgenic animals are designed to help researchers understand gene regulation and its impact on normal physiological functions and development.

[New NCERT Class 12th Page No. 183]

41. **(2)**

- For efficient cloning, a vector should have very few, preferably a single recognition site for a restriction enzyme.
- This is important because it ensures that the vector can be cut at a specific, controlled location, allowing for the precise insertion of the alien DNA without disrupting other essential functions of the vector.

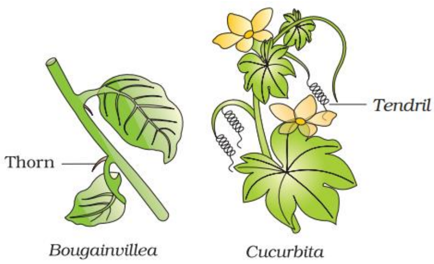
[New NCERT Class 12th Page No. 169]

42. **(1)**

Primers are small chemically synthesised oligonucleotides and complementary to the template DNA.

[New NCERT Class 12th Page No. 173]

43. **(2)**



The thorn and tendrils of *Bougainvillea* and *Cucurbita* represent homology and are result of divergent evolution.

[New NCERT Class 12th Page No. 115]

44. **(2)**

- Restriction enzymes belong to a large class of enzymes called nucleases.
- *Hind II* cuts DNA at a specific sequence of six base pairs.
- DNA molecules are negatively charged.
- Restriction enzymes cut the sugar-phosphate backbone of DNA.

[New NCERT Class 12th Page No. 165, 166, 168]

45. **(2)**

- *Bt* cotton contains the *cry* gene from *Bacillus thuringiensis* bacteria, which produces a protein toxic to insects like cotton bollworms.
- The insecticidal protein kills insects by disrupting their gut cells.

[New NCERT Class 12th Page No. 180]

46. **(1)**

Ramapithecus was more man-like while *Dryopithecus* was more ape-like.

[New NCERT Class 12th Page No. 124]

47. **(4)**

- Saltation is a single step large mutation.
- The amphibians evolved into reptiles.
- Branching descent and natural selection are two key concepts of Darwinian theory of evolution.
- Sea weeds and few plants existed probably around 320 mya.
- About 65 mya, the dinosaurs suddenly disappeared from the earth.

[New NCERT Class 12th Page No. 114]

48. **(4)**

Pst I	amp^R
Endonuclease	Cleaves at specific position within the linear DNA.
Cellulase	Degradation of plant cell wall
PCR	Amplification of genetic material

[New NCERT Class 12th Page No. 166, 169, 171, 172]